

Heart Disease Diagnosis Prediction

1. Introduction:

The goal of this project is to create and assess machine learning models for heart disease prediction. The objective is to develop models that can identify people who are more likely to develop heart disease by utilizing a dataset that includes a variety of health indicators. This could help with early intervention and individualized healthcare plans.

2. Dataset Exploration And Analysis:

To comprehend the dataset's properties and the connections between features and the target variable, exploratory data analysis, or EDA, was carried out.

Numerical Feature Analysis: Box plots were utilized to illustrate the distribution of numerical features in relation to the target variable, and summary statistics were produced for each feature.

Categorical Feature Analysis: The distribution of categorical features in connection to the existence or absence of heart disease was visualized using count plots.

Correlation Analysis: To show the pairwise correlations between every feature, including the target variable, a correlation heatmap was created. This made it easier to find characteristics that were strongly associated with heart disease, either positively or negatively.

Distribution Analysis: To see the distributions of numerical features, histograms were plotted.

3. Data Preprocessing:

Scaling the features and managing duplicate values and outliers were part of the data preprocessing process.

Outlier Handling: To find and limit outliers in numerical features such as chol, trestbps, thalach, and oldpeak, the Interquartile Range (IQR) approach was employed.

Duplicate Removal: To guarantee data integrity, duplicate rows in the dataset were found and eliminated.

Feature Scaling: Training, validation, and test sets were transformed after StandardScaler was applied to the training data. For models that are sensitive to

feature scales, it is crucial that the features be standardized to have a zero mean and unit variance.

4. Model Training:

The ability of a few machine learning models to predict heart disease was assessed after they were trained. Training (70%), validation (15%), and testing (15%) sets were created from the dataset.

Feature Selection: Eight features were chosen as a subset of the most significant features using Recursive Feature Elimination (RFE) with a RandomForestClassifier estimator. The models were then trained using the chosen features.

Trained Models: The following were models were trained and optimized:

- Support Vector Machine (SVM)

- Classifier Gradient Boosting Classifier
- Neural Network (MLPClassifier)

5. Model Evaluation:

The models were evaluated using the following metrics:

- ✓ Accuracy
- ✓ F1 Score
- ✓ AUC-ROC (Area Under the Receiver Operating Characteristic Curve)
- ✓ Classification Report (including precision, recall, and F1-score for each class)
- ✓ Confusion Matrix

6. Results:

The selected metrics were used to assess each model's performance on the test set. To visually compare each model's performance across various classification thresholds, the ROC curves for each model were plotted. For every model, confusion matrices displaying the quantity of true positives, true negatives, false positives, and false negatives were created.

7. Conclusion:

People who are at a higher risk of heart disease can be identified with the help of predictive models. By promoting early detection, these models allow for prompt interventions that may stop or postpone the onset of heart disease. The outcomes of predictions can help prioritize patients who are more at risk and facilitate the effective distribution of healthcare resources. Customizing treatment regimens and

lifestyle advice is made possible by an understanding of the unique risk factors that the models highlight. Better clinical judgment and possibly better patient health outcomes can result from the use of predictive analytics. The models offer more data-driven insights that can enhance clinicians' knowledge.